Home Assignment 2

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Q. 1: 1. All the movies whose rank is at least 9.

a. Using “rank” as a name of a column in MySql is problematic. Explain why and explain how you cope with it.

(Hint: There are 49,573 movies whose rank is at least 5. Check your query by adapting it to this value and see that you get the same result.)

A.1:

a) According to the MySQL documentation of keywords and reserved words, using ‘rank’ as a column name is problematic as it also serves as a keyword.

2 possible fixes to this limitation are:

1. to wrap the column name in backsticks (as displayed in the query adjusted for the hint).
2. Specify ‘rank’ refers to a column from a table by adding the table name before the keyword as displayed in the query.

Query: SELECT * FROM movies

WHERE movies.rank >=9;

Rows returned: 1353

Adjusted query for movies with rank

Higher than 5:

Query: SELECT * FROM.movies

WHERE `rank` >=5;

Rows returned: 49,573

id	name	year	rank
18	$40,000	1996	9.6
330	+1 -1	1987	9.6
837	1001 Nights	1998	9
966	12 (2003/II)	2003	9.8
995	12 stulyev	1971	9.3
1100	14 Million Dreams	2003	9.5
1595	2+1	2004	9.2
1847	24	2001	9
2057	2wks, 1yr	2002	9.4
2099	3 Feet Under: Digging Deep for the Geoduck	2003	9
2352	36K	2000	9.5
2656	5 Card Stud	2002	9.4
2998	7 Songs	2003	9
3112	8 $	1999	9.2
3421	A La Cart	1997	9
4458	Aatomik	1970	9
4552	Abang	1981	9.3
4702	Abduction of Figaro	1984	9.1
4721	Abdulladzhan, ili posvyashchayetsya Stiven…	1991	9.1
4948	Able's House Is Green, The	2003	9.4
5086	Abrahams Gold	1990	9.3
5496	Accordon	2004	9.7
5513	Accueil familial, vieillir comme chez soi, L'	1996	9
5514	Accueil temporaire des personnes ges, ou l…	1994	9
5734	Aclik	1974	9.4
5966	Actiunea Autobuzul	1978	9.4
6202	Adamah	1947	9.4
6220	Adaptatziya	1981	9.7

Q. 2: All the different role names that include the string ‘him’

i. Take care of being case insensitive

ii. Note that the same role might appear in multiple movies, yet should only appear once in the results (Hint: There are 46,686 roles that contain the string ‘her’. Check your query by adapting it to this value and see that you get the same result.)

iii. Do you find the role name ‘Himself’ appropriate? To which problems it might lead? Explain why.

iv. Suggest a way to improve the returned list.

A. 2:

Part 1: I made sure the query was case insensitive by using lower() in the where statement.

part 2: To prevent the same role appearing multiple times, I selected only distinct roles.

Query:

```
select distinct `role` from imdb_ijs.roles

where lower(`role`) like '%him%';
```

Rows returned: 15,524

role
Himself
Himself (DJ, New York)
Himself/Disc Jockey
Himself (1997)
Himself (1998-1999)
Himself (1999-2000)
Himself (Nation of Islam)
Himself (Outkast)
Himself - Presenter
Himself - Performer
Performer/Himself
Himself/Performer

Part 3: I think the role name ‘himself’ isn’t suitable as it doesn’t supply any information about who is portrayed, moreover several ‘himself’ portrayals may still serve different roles. I think it may lead to problems when filtering out unique roles since some actors portray themselves in different movies.

Part 4: To bridge this gap, I chose to create a unified column using the format “actor_name as himself” for an easy link by selecting the corresponding columns from their respective tables and filtering based on ID as a key. On top of that I chose to select and filter the name column of corresponding movies for some added context.

Query:

```
select a.id as actor_id,
concat(a.first_name,' ', a.last_name,' as ', r.role) as 'actor/role',
m.name
from roles as r, actors as a, movies as m
where r.actor_id = a.id
and r.movie_id = m.id
and lower(r.`role`) LIKE '%himself%'
order by a.id;
```

Rows returned: 119,997

actor_id	actor/role	name
5	Antonio 'El de Chipiona' as Himself	Guitarra muda, La
17	Luis Roberto 'Formiga' as Himself	OP Pro 88 - Barra, Rio
18	Murray the 'K' as Himself (DJ, New York)	Beatles: The First U.S. Visit, The
18	Murray the 'K' as Himself/Disc Jockey	Day the Music Died, The
18	Murray the 'K' as Himself	I Wanna Hold Your Hand
18	Murray the 'K' as Himself	"Beatles Anthology, The"
26	Izzy 'The Religious Nut' as Himself	Twister: A Musical Catastrophe
28	Oswald 'Ô' as Himself	Skate - O Esporte Emoo
34	Shaggy 2 Dope as Himself	Juggalo Championship Wrestling Volume 1
34	Shaggy 2 Dope as Himself	Summerslam
34	Shaggy 2 Dope as Himself	WCW Road Wild '99
34	Shaggy 2 Dope as Himself (1997)	"Raw Is War"
34	Shaggy 2 Dope as Himself (1998-1999)	"Sunday Night Heat"
34	Shaggy 2 Dope as Himself (1999-2000)	"WCW Monday Nitro"
34	Shaggy 2 Dope as Himself (1998-1999)	"WCW Saturday Night"
34	Shaggy 2 Dope as Himself (1998-1999)	"WCW Thunder"
34	Shaggy 2 Dope as Himself (1998-1999)	"WCW Worldwide Wrestling"
35	Mode 2 as Himself	Just to Get a Rep

Q.3: All movies whose name is longer than 95 characters, ordered by length

a. Explain the prevalence of names in each length. Why is certain length much more common than the others?

b. Bonus: Suggest a way to identify some of the problematic names and implement it.

A.3:

a. I used length() to measure the length of movie names for filtration and ordering, besides that I also added the name length column to check for name length prevalence.

For name length of 97-99 is seemingly a single movie from 1900 registered in 4 installments. For movies with a name length of 100, many seem to be in foreign languages (mainly German, French and Spanish).

The most prevalent name length above 95 is 100 characters. As can be seen in the IMDB database creation script, this is caused by the character limit being set to 100, resulting in longer movie names being cut off at the hundredth character.

```
DROP TABLE IF EXISTS `movies`;
CREATE TABLE `movies` (
  `id` int(11) NOT NULL DEFAULT 0,
  `name` varchar(100) DEFAULT NULL,
  `year` int(11) DEFAULT NULL,
  `rank` float DEFAULT NULL,
  PRIMARY KEY (`id`),
  KEY `movies_name` (`name`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3 COLLATE=utf8mb3_general_
```

Query:

select *, length(name) as name_length

from movies

where length(name)>95

order by length(name) desc;

Rows returned: 104

id	name	year	rank	name...
174	'N herinnering wijlen Z.M. Willem III een rijtoer makende door het Vondelpark te Amsterdam, 'N (1899	1899	NULL	100
999	12 vistas tomadas 6 horas despues de la catastrofe del ferrocarril Mexicano en el puente de Metlac (	1905	NULL	100
2028	29th Telluride Film Festival Aug. 30 - Sept. 2, 2002: Terry Gilliam Interviewed by Salman Rushdie (2	2003	NULL	100
6681	Adrienne Clarkson Presents: A Tribute to Peppiatt & Aylesworth: Canada's First Television Comedy Tea	1996	NULL	100
6905	Adventures of a Man in Search of a Heart: A Joleron Production Starring the Tin Woodman from the Lan	1984	NULL	100
24045	Auf rauschender Fahrt - Ein Segelsportfilm und was Operateur Huckebein bei seiner Aufnahme erlebte (	1922	NULL	100
24100	Aufstieg - Ein Film vom Werden und sozialen Wirken des Verbandes der Fabrikarbeiter Deutschlands (19	1929	NULL	100
47038	Brigitte, Laura, Ursula, Monica, Raquel, Litz, Florinda, Barbara, Claudia, e Sofia le chiamo tutte..	1974	NULL	100
52626	Calles de Orizaba, rapidos de Rio Blanco, instalacin electrica, laguna de nogales, y cerveceria Moct	1908	NULL	100
53384	Canada Vignettes: Don Messer - His Land and His Music - Charlie Chamberlain 1911-1972, Marg Osburne	1979	NULL	100
64306	Cinma de notre temps: Le dinosaure et le bbe, dialogue en huit parties entre Fritz Lang et Jean-Luc	1967	NULL	100
64310	Cinma de notre temps: Portrait de Michel Simon par Jean Renoir ou Portrait de Jean Renoir par Michel	1966	NULL	100
64479	Circular Panoramic View of St. Pierre from the Lighthouse, Showing Mt. Pelee Smoking in the Distance	1902	NULL	100
64480	Circular Panoramic View of St. Pierre, Showing the Ruins of the Catholic Cathedral and Mt. Pelee Smo	1902	NULL	100

b. One common trait of incomplete names (due to supposed character limit) is that they end with either an open parentheses, at times there is content and at times not.

To identify such problematic names I chose to filter out movie names that contain an opening parentheses, then use right() to filter out names with unclosed parentheses which would point to an incomplete name.

Query:

```
select * from movies

where length(name) = 100

and name like '%(%'

and right(`name`,1) != ')';
```

rows returned: 34

id	name	year	rank
174	'N herinnering wijlen Z.M. Willem III een rijtoer makende door het Vondelpark te Amsterdam, 'N (1899	1899	NULL
999	12 vistas tomadas 6 horas despues de la catastrofe del ferrocarril Mexicano en el puente de Metlac (	1905	NULL
2028	29th Telluride Film Festival Aug. 30 - Sept. 2, 2002: Terry Gilliam Interviewed by Salman Rushdie (2	2003	NULL
24045	Auf rauschender Fahrt - Ein Segelsportfilm und was Operateur Huckebein bei seiner Aufnahme erlebte (	1922	NULL
24100	Aufstieg - Ein Film vom Werden und sozialen Wirken des Verbandes der Fabrikarbeiter Deutschlands (19	1929	NULL
71204	Cortejo Eucarstico Saindo da S do Porto no Aniversrio da Sagrao do Eminentssimo Cardeal Amrico (1896	1896	NULL
126357	Geschichte von Herrn Zett, der hinaufstieg sein Leben zu beenden und von Anton, dem Angler, Die (199	1997	NULL
149509	Hora de los hornos: Notas y testimonios sobre el neocolonialismo, la violencia y la liberacin, La (1	1968	8.2
151626	How to Make Dhyrak: A Dramatic Work for Three Players and Camera, Truncated with Only Two Players (1	1999	NULL
154288	I Killed My Lesbian Wife, Hung Her on a Meat Hook, and Now I Have a Three-Picture Deal at Disney (19	1993	4.5
157133	Imperfect Union: Canadian Labour and the Left, Part 1: International Background - Canadian Roots (19	1989	NULL
167207	Jeunes gens du Stade Montois s'entrainant la course ou au saut en vue de championats au Waux-Hall (	1904	NULL
186096	Launching of the New Battleship 'Ohio' at San Francisco, Cal. When President McKinley Was There (190	1901	NULL
203665	Man with the Smallest Penis in Existence and the Electron Microscope Technician Who Loved Him, The (	2003	5.4

Q. 4: Find at least 3 first names in the actors table that are most likely to be mistakes. There are many ways to make mistakes so we are liberal and will accept any justified answer.

i. Explain how you found each of the names

ii. Suggest a possible cause of the mistake

iii. Suggest a way that would prevent the problem in the first place or identify it afterwards. Bonus: Find mistakes where it is not easy to do so.

A.4:

Below are the query and likely mistakes I found:

```
select first_name, last_name from actors

where first_name regexp '[^\\p{L} ()''.0-9-]'

or first_name like ' %and%';
```

first_name	last_name
Allen Fontenot & The	Country Cajuns
Kool & The	Gang
L:Ron	Harald
Johan H:son	Kjellgren
Curt H:son	Nilsson
Luca & Sebastian	Punzo
Greg 'M&M'	Werts
!Nqate	Xqamxebe
Nicaela & Shelbie	Weigel

Rows returned: 9

After googling each one, I concluded that names using the character '&' are likely either a band, or a pair of twins. In the case of twins, I believe this conclusion justifies classifying such names as mistakes, since they don't credit both actors individually.

(Note: the names containing an exclamation point and colons aren't mistakes, but may be in the future).

I think such mistakes may arise from either from lumping two actors together when manually crediting a movie because of laziness or for the sake of row count reduction. Moreover, such mistakes may arise from automated parsing errors if IMDB scrubs data from text files – for example movie credits for a single role played by both twins.

I think this mistake could have been prevented in the first place if the use of Ampersands (and perhaps also the word 'and') was monitored by a human or even by a program checking if the name refers to a single entity, which can in most cases be determined with a simple google search. Based on the program's/manual search, such mistakes can be split into 2 rows.

To easily detect and verify suspicious names, I suggest the above regexp pattern, which will detect any name not containing any letter character from any language, parentheses, quotation marks, dots, numbers or dashes.

However, note this isn't a perfect solution considering the pattern also flagged 4 names which turned out not to be mistakes..